

Grade 8
Unit 8
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

The student council at Julian's school is planning a dance. Their costs will be \$44.50 for decorations, \$60 for the DJ, and \$2.50 for snacks per person who attends. They are selling tickets for \$5.25 each. The equation $44.50 + 60 + 2.50t = 5.25t$, where t is the number of tickets they sell, can be used to determine the number of tickets they need to sell to "break even," or have their revenue equal their costs.

1. Solve the equation $44.50 + 60 + 2.50t = 5.25t$.

$$\begin{aligned} 104.50 + 2.50t &= 5.25t \\ -2.50t &- 2.50t \\ 104.50 &= 2.75t \\ 38 &= t \end{aligned}$$

2. What does the value of t that you determined in question 1 represent in the context?

38 tickets must be sold to break even.

At an airport, one airplane is taking off and another one is descending for landing. The airplane taking off starts at an altitude of 200 feet and ascends at a rate of 2,530 feet per minute. The airplane landing starts at an altitude of 33,400 feet and descends at a rate of 1,620 feet per minute. The equation $200 + 2,530m = 33,400 - 1,620m$, where m is the number of minutes, can be used to determine when the two planes will have the same altitude.

3. Solve the equation $200 + 2,530m = 33,400 - 1,620m$.

$$\begin{aligned} +1620m &+ 1620m \\ 200 + 4150m &= 33,400 \\ -200 &- 200 \\ 4150m &= 33,200 \\ m &= 8 \end{aligned}$$

4. After how many minutes will the planes be at the same altitude?

8 minutes

Mia is considering two catering companies for a party. Daisy's Meals charges \$31.75 per person, while Clearwater Catering charges \$26.50 per person with an additional administrative fee of \$47.25.

5. Solve the equation $31.75p = 26.50p + 47.25$, where p is the number of people, to determine at what number of people both companies will charge the same amount.

$$31.75p = 26.50p + 47.25$$

$$5.25p = 47.25$$

$$p = 9$$

6. What does the value of p that you found in question 5 represent in the context?
For 9 people, both companies charge the same.

7. Check your work by finding the total cost of both catering companies.

Cost at Daisy's Meals	Cost at Clearwater Catering
$31.75p$ $31.75(9) = 287.75$	$26.50p + 47.25$ $26.50(9) + 47.25 = 285.75$

Orion is comparing his accounts. He has \$360 in his savings account and deposits \$35 per week. He has \$2,180 in his checking account and withdraws \$95 per week.

8. Solve the equation $360 + 35w = 2,180 - 95w$, where w is the number of weeks, to determine when the two accounts will have the same amount.

$$360 + 35w = 2180 - 95w$$

$$360 + 130w = 2180$$

$$130w = 1820$$

$$w = 14.0$$

9. What does the value of w that you found in question 8 represent in the context?
At 14 weeks, the balance will be the same in both accounts.

10. Check your work by finding the amount in each account.

Amount in Savings Account	Amount in Checking Account
$360 + 35w$ $360 + 35 \cdot 14 = 850$	$2,180 - 95w$ $2180 - 95(14) = 850$